BME 6938 · ADVANCED BIOMEDICAL AI · SPRING 2027

LECTURE 1-1

Course Logistics

Welcome to Advanced Biomedical AI



Parisa Rashidi · Herbert Wertheim College of Engineering, University of Florida

Why This Course?

Biomedical AI has moved from one model per task to large pretrained models adapted to many tasks.

Current research

Recent papers every week

Foundation and language models

Agents and generative AI

Critical reading

Present and critique papers

Separate what is shown

from what is claimed

Your own study

A research paper on an

open problem you choose

with a reproducible repository

Builds on BME 6938 Medical AI (Fundamentals of Biomedical AI) or an equivalent ML or deep learning course

Time and Location

Tuesday

1:55 to 2:45 PM

Period 7 · Lecture

Thursday

1:55 to 3:50 PM

Periods 7 and 8 · Two paper sessions

Black Hall, room BLK 0315 · in person · laptop required

Spring break: March 13 to 20 · Last day of classes: April 21



Instructor



Dr. Parisa Rashidi

Biomedical Engineering

Email parisa.rashidi@ufl.edu

Office hours By appointment on Zoom

Research: multimodal AI, agentic systems, and clinical decision support

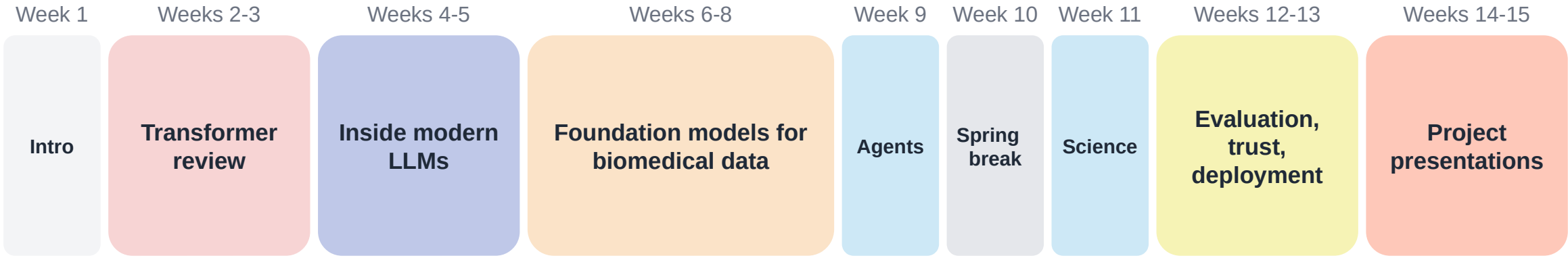
Course Objectives

By the end of the semester you should be able to:

1. Explain the principles behind current model architectures and training paradigms in biomedical AI.
2. Critically read, present, and critique current biomedical AI research papers.
3. Formulate a problem and select appropriate data, models, and methods to address it.
4. Evaluate biomedical AI models rigorously, beyond aggregate accuracy.
5. Identify the requirements for translating AI models into biomedical and clinical use.
6. Design, carry out, and report a research project in biomedical AI.

Course Map

Fifteen weeks. The last two weeks are reserved for project presentations.



2 Transformer review, Part 1

3 Transformer review, Part 2

4 Inside modern LLMs: 2026 architectures

5 Post-training, reasoning, and LoRA

6 Self-supervised learning and imaging FMs

7 Vision-language and multimodal models

8 EHR and physiological time series

9 Retrieval, tools, and agents

11 Generative models and AI for science

12 Evaluation, robustness, and fairness

13 Deployment and human-AI teaming

A Typical Week

Tuesday → **Before Thursday** → **Thursday**

Lecture (50 min)
Concepts, animation clips, and a notebook

Read both papers
Reading responses on Canvas by 1:00 PM Thursday

Period 7: paper 1
Period 8: paper 2
Students lead both

Each Thursday paper session (50 minutes)





Grading and Assignments

Grading

1,000 points in three parts.



Paper presentations and critiques

2 presentations and 2 critiques

100 points each, rubric on Canvas

Reading responses and participation

Reading responses (1 point each, graded automatically), proposal peer review, participation

Term project

Proposal 5% · progress report 5%

presentation 10% · paper and repo 20%

Late work: 5% deducted per late day unless the assignment states otherwise

Paper Sessions

Every student presents twice and critiques twice, in groups.

Presentation group · 20 min

Explain the problem and why it matters

Data, model, and key design choices

Results: what they demonstrate

Strengths relative to prior work

Clear slides, balanced speaking time

Critique group · 15 min

Restate the core claims accurately

Critique methods and validation

Do the results support the claims?

Limitations, biases, and ethics

Constructive suggestions

Graded by the instructor · all group members receive the group grade · assignments on Canvas by Jan 15

Reading Responses

For each Thursday paper you are not presenting or critiquing.

1. One paragraph (about 150 words): the main claim of the paper and whether the evidence supports it.

2. One discussion question you want the class to address.

1 point each, graded automatically on submission · locks at 1:00 PM Thursday · lowest two dropped



The Term Project

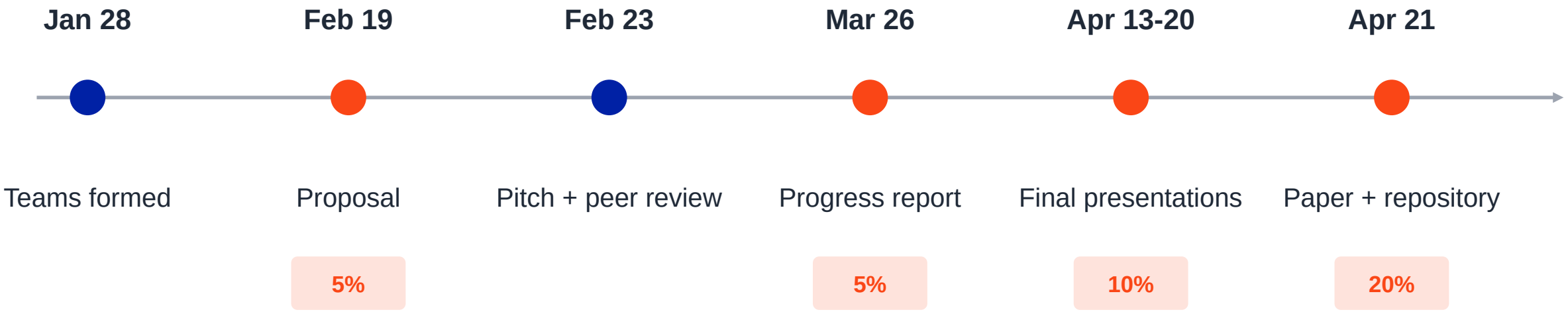
Term Project: A Research Paper

Teams of three choose an open problem, run an original study, and write it up as a paper.

A baseline	Compare against an appropriate, honest baseline
Rigorous evaluation	Beyond one aggregate metric: calibration, subgroups, or a shift or robustness test
Failure analysis	Where and why does the method fail?
Reproducibility	4-page conference-style paper and a GitHub repository that runs from its README

No topic yet? Suggested problems with open datasets are listed on the course website.

Project Milestones



Final talk: 12 min + 4 min questions · all slides due before Tue, Apr 13

Start Data Access Now

Credentialed clinical datasets take weeks to unlock.

Credentialed (start this week)

MIMIC-IV, MIMIC-CXR, eICU

CITI training + PhysioNet credentialing

Can take several weeks

Open (no credentialing)

PTB-XL (ECG)

MedMNIST (imaging)

TCGA (pathology, genomics)

MedQA, PubMedQA (text)

Never commit patient data to a repository or paste credentialed data into online AI services.



Resources and Policies

Where Things Are

Canvas

- Announcements
- Assignments and submissions
- Grades
- Paper assignments
- Turn on notifications

Course GitHub repository

- Slides and animation clips
- Notebooks (Colab and HiPerGator)
- Paper list and rubrics
- Project templates

If Canvas and any other source disagree, Canvas is correct.

Computing

Every notebook runs on Google Colab and on HiPerGator.

Google Colab	HiPerGator	Software
Free GPU for lecture notebooks	UF's supercomputer	Python, PyTorch,
Nothing to install	GPUs for project work	Hugging Face, scikit-learn
Session limits on the free tier	Course allocation (details on Canvas)	GitHub for code

Lecture notebooks are ungraded reference code that follows each lecture step by step

Getting Help

Three steps, in order.

1

Try it yourself

Slides, notes, notebooks

Documentation and papers

AI tools, used responsibly

2

Ask your peers

Canvas discussion board

Say what you tried

Help others too

3

Contact the instructor

Email for a Zoom appointment

Include details and

what you already tried

Using AI Tools

This is an AI course: use AI tools, and use them well.

Encouraged

Understanding papers and concepts

Coding help and debugging

Editing your own writing

You own every claim and line of code

Not allowed

Reading responses or critiques you did not write from your own reading

Patient or credentialed data in online AI services

Unverified citations

Project papers include a short statement of how AI tools were used.

Course Policies

Attendance	Required; paper sessions depend on everyone being there
Your own work	Write your own code and text; plagiarism is checked
Group work	Contribute equally; speaking time is shared
Late work	5% per late day unless stated otherwise
University policies	Linked in the syllabus: go.ufl.edu/syllabuspolicies

This Week

Thu, Jan 14	Read Moor et al., "Foundation models for generalist medical AI" (Nature 2023)
Fri, Jan 15	Paper and group assignments posted on Canvas
Tue, Jan 19	Transformer Review, Part 1 (notebook posted as reference code)
This week	Introduce yourself on the Canvas discussion board; start CITI training if you may use MIMIC

Introduce Yourself

On the Canvas discussion board, and briefly now.

Your name and program

Your research area

One biomedical AI problem you care about

What you want from this course